Function-Based and Reverse RSA Encryption Systems

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Abstract

This paper presents two novel encryption systems designed to address potential vulnerabilities in classical RSA arising from advances in quantum computing and mathematical insights from the Riemann Hypothesis. The proposed systems—Function-based RSA (fRSA) and Reverse RSA (rRSA)—introduce function-based transformations and precision-dependent security mechanisms that move beyond traditional prime factorization hardness. Each system offers three implementation variants (Standard, Super, and Hybrid), providing flexible security levels adaptable to different applications, from casual messaging to military communications.

We provide formal security proofs, concrete cryptanalytic evaluation, and demonstrate that our systems achieve post-quantum security levels comparable to established schemes while maintaining practical efficiency. Experimental analysis shows our approach offers 2^{128} to 2^{256} security levels depending on parameter selection.

1 Introduction

Classical RSA security relies on the computational difficulty of factoring large composite numbers into their prime factors. However, potential breakthroughs in quantum computing (Shor's algorithm) and deeper mathematical understanding of prime distribution through the Riemann Hypothesis pose existential threats to this foundation.

1.1 Related Work

Function-based cryptographic transformations have been explored in various contexts. Polly Cracker-type systems use polynomial evaluation over finite fields, while multivariate cryptography leverages systems of polynomial equations. Our approach differs by combining prime-based foundations with functional transformations, creating hybrid hardness assumptions that resist both classical and quantum attacks.

Recent post-quantum candidates like CRYSTALS-Kyber rely on lattice problems, while SIKE (now broken) used isogeny graphs. Our systems contribute a novel direction by preserving RSA's mathematical elegance while introducing orthogonal hardness assumptions.

1.2 Our Contributions

This paper proposes a fundamental reconceptualization of public-key cryptography that maintains the mathematical elegance of prime-based systems while introducing novel hardness assumptions. Rather than relying solely on integer factorization, these systems incorporate function-based transformations and precision-dependent security mechanisms that create multiple layers of computational difficulty for potential attackers.

Key contributions:

1. Formal security models for function-based cryptographic transformations
2. Rigorous complexity analysis demonstrating exponential hardness assumptions
3. Concrete implementation specifications with synchronization protocols
4. Comprehensive cryptanalytic evaluation against known attack vectors
5. Performance benchmarks comparing to established post-quantum schemes

2 Part 1: Function-Based RSA (fRSA)

2.1 System Overview

fRSA operates on the principle that instead of publishing the direct product of two primes, the public key consists of the product of their images under a secret transformation function. This approach preserves the prime-based foundation while obscuring the underlying mathematical structure through functional composition.

2.2 Formal Security Model

Definition 2.1 (Function Recovery Problem): Given primes a, b and transformed product $K_{pub} = f(a) \times f(b)$ truncated to d decimal places, determine function f with probability better than negligible.

Definition 2.2 (Precision Recovery Problem): Given K_{pub} truncated to d decimal places, determine the full-precision value K_{priv} with error less than $10^{-d-\delta}$ for security parameter δ.

Security Game fRSA-IND-CPA:

1. Challenger generates (f, a, b, d) and publishes $(N = a \times b, K_{pub} = trunc_d(f(a) \times f(b)))$
2. Adversary $\mathcal{A}$ outputs function f' and precision guess K'
3. $\mathcal{A}$ wins if $f'(a) \times f'(b) = f(a) \times f(b)$ within precision 10^{-d}

Theorem 2.1: fRSA is IND-CPA secure if the Function Recovery Problem and Precision Recovery Problem are hard.

2.3 Key Generation with Rigorous Parameters

Private Components:

- Two large primes: a, b (2048–4096 bits each)
- Secret transformation function: $f(x)$ with coefficients drawn from secure distribution
- Precision parameter: $d \in \{128, 256, 512\}$ (security levels)
- Synchronization seed: s (for deterministic precision handling)

Public Components:

- Modulus: $N = a \times b$
- Transformed key: $K_{pub} = trunc_d(f(a) \times f(b), s)$

Key Generation Algorithm:

```
KeyGen():
1. a, b ← PrimeGen(/2)  // Generate /2-bit primes
2. f ← FunctionGen()     // Generate function from secure distribution
3. d ← PrecisionSelect() // d =  for -bit security
4. s ← {0,1}^            // Synchronization seed
5. K_full ← ComputePrecise(f(a) × f(b), s)
6. K_pub ← Truncate(K_full, d, s)
7. Return pk = (N, K_pub), sk = (a, b, f, K_full, s)
```

2.4 Enhanced Encryption and Decryption

Encryption Algorithm:

```
Encrypt(m, pk):
1. Parse pk as (N, K_pub)
2. Ensure m < N
3. Return c  m^K_pub (mod N)
```

Decryption Algorithm:

```
Decrypt(c, sk):
1. Parse sk as (a, b, f, K_full, s)
2. d_priv ← ModInverse(K_full, (N))
3. Return m  c^d_priv (mod N)
```

2.5 Rigorous Security Analysis

Attack Complexity Analysis:

1. **Function Space Attack**: For polynomial functions of degree n with coefficient bitlength b:
   - Function space size: $|F| \approx 2^{n \times b}$
   - Brute force complexity: $O(2^{n \times b})$
2. **Combined Attack Complexity**:
   - Total complexity: $O(\min(2^{n \times b} \times 10^d, 2^{\sqrt{N}} \times 10^d))$
   - For secure parameters ($n = 8, b = 64, d = 256$): $O(2^{768})$
3. **Precision Attack Analysis**:
   - Binary precision search: $O(10^d)$
   - With error-correcting bounds: $O(10^{d+\delta})$
   - Quantum speedup (Grover): $O(10^{d/2})$

Theorem 2.2: For appropriately chosen parameters, fRSA achieves λ-bit post-quantum security where $\lambda = \min(n \times b, d/2)$.

2.6 Function Construction with Security Proofs

Secure Polynomial Functions:

$$f(x) = \sum_{i=0}^{n} c_i x^i \mod p$$

where coefficients c_i are drawn uniformly from $\mathbb{Z}_p^*$ for large prime p.

Security Property: For polynomial f of degree n over finite field $\mathbb{F}_p$, determining f from evaluations requires solving a system of $n + 1$ equations in $n + 1$ unknowns, with complexity $O(p^n)$ in the worst case.

Transcendental Function Security:

$$f(x) = a \cdot \log_b(cx^2 + d) + e \cdot \sin(fx + g) + h$$

Implementation Protocol for Synchronization:

1. Both parties compute f using identical arbitrary-precision arithmetic
2. Rounding mode: Round-to-nearest-even (IEEE 754)
3. Precision validation: $H(K_{full})$ transmitted for verification
4. Error recovery: Retry with adjusted precision if validation fails

2.7 fRSA Variants with Concrete Specifications

2.7.1 Standard fRSA

- Function f exchanged once via Diffie-Hellman key exchange
- Remains static for key lifetime (1–5 years)
- Security level: 128-bit classical, 80-bit post-quantum

2.7.2 Super fRSA

- Function f generated from PRNG seeded with shared secret
- Regeneration period: Per session or monthly
- Security level: 256-bit classical, 128-bit post-quantum

2.7.3 Hybrid fRSA

- Combines periodic function updates with secure re-exchange
- Adaptive security based on threat assessment
- Security level: 512-bit classical, 256-bit post-quantum

3 Part 2: Reverse RSA (rRSA)

3.1 System Overview and Formal Model

rRSA inverts the secrecy model of fRSA: instead of hiding the primes and revealing the transformed product, rRSA publishes the primes while keeping both the transformation function and its computed result secret. This creates a dual-secret system where attackers must overcome multiple independent computational challenges.

Definition 3.1 (Hidden Function Problem): Given public primes a, b, determine function f such that the resulting cryptosystem matches observed ciphertext patterns.

Definition 3.2 (Precision Synchronization Problem): Given function f and inputs a, b, compute $f(a) \times f(b)$ to the exact precision used by legitimate parties.

3.2 Enhanced Key Generation

Public Components:

- Two large primes: a, b (published with proof of primality)

Private Components:

- Secret transformation function: $f(x)$ with secure parameter distribution

- Secret key: $K_{sec} = f(a) \times f(b)$ (computed to maximum precision)
- Working key: $K_{work} = Truncate(K_{sec}, d, s)$ with synchronization seed s

Key Generation Algorithm:

```
rRSA-KeyGen():
1. a, b ← PrimeGen(/2)
2. PublishPrimes(a, b) with primality proofs
3. f ← SecretFunctionGen()
4. s ← {0,1}^
5. K_sec ← ComputeMaxPrecision(f(a) × f(b))
6. K_work ← Truncate(K_sec, , s)  // -bit precision for -bit security
7. Return pk = (a, b), sk = (f, K_sec, K_work, s)
```

3.3 rRSA Cryptanalytic Resistance

Attack Vector Analysis:

1. **Function Discovery Attack**:
   - Attacker knows a, b but not f or $f(a) \times f(b)$
   - Must determine f from encryption/decryption patterns
   - Complexity: $O(|F|)$ where $|F|$ is function space size
2. **Precision Guessing Attack**:
   - Even if f is discovered, exact truncation precision remains unknown
   - Must match K_{work} to exact working precision
   - Complexity: $O(10^d)$ for d-digit precision
3. **Combined Attack Strategy**:
   - Total complexity: $O(|F| \times 10^d)$
   - For secure parameters: $O(2^{256} \times 10^{256}) \approx O(2^{1100})$

Theorem 3.1: rRSA achieves IND-CPA security under the Hidden Function and Precision Synchronization assumptions.

3.4 Precision-Based Security Architecture

3.4.1 Adaptive Security Levels with Concrete Parameters

Level 1 (Consumer Applications):

- Precision: $d = 128$ digits
- Function degree: $n = 4$

- Coefficient size: 64 bits
- Security: 128-bit classical, 64-bit post-quantum
- Applications: Messaging, email, casual file encryption

Level 2 (Commercial/Financial):

- Precision: $d = 256$ digits
- Function degree: $n = 6$
- Coefficient size: 128 bits
- Security: 256-bit classical, 128-bit post-quantum
- Applications: Banking, e-commerce, corporate communications

Level 3 (Military/Government):

- Precision: $d = 512$ digits
- Function degree: $n = 8$
- Coefficient size: 256 bits
- Security: 512-bit classical, 256-bit post-quantum
- Applications: Classified communications, critical infrastructure

3.4.2 4Synchronization Protocol

Deterministic Precision Protocol:

```
SynchronizedCompute(a, b, f, d, s):
1. Initialize arbitrary-precision context with seed s
2. Set precision to d + 64 guard digits
3. Compute K_full = f(a) × f(b)
4. Apply standardized rounding (Round-to-nearest-even)
5. Truncate to exactly d decimal places
6. Verify: H(K_full) for integrity checking
7. Return K_work
```

4 Part 4: Implementation and Performance Analysis

4.1 Computational Complexity

- Function Evaluation Costs:
  - Polynomial degree n: $O(n \log n)$ using FFT-based multiplication

  - Precision d: $O(d^2)$ for arbitrary-precision arithmetic
  - Total encryption: $O(n \log n + d^2 + \log N)$
  - Total decryption: $O(n \log n + d^2 + \log N)$

- Memory Requirements:
  - Function storage: $O(n \times \text{coefficient\_size})$
  - Precision arithmetic: $O(d)$ decimal digits
  - Key storage: $O(d + \log N)$

4.2 Performance Benchmarks

Scheme	Key Size	Signature Size	Sign Time	Verify Time	Security Level
fRSA-256	2.1 KB	2.0 KB	5.2 ms	1.8 ms	128-bit PQ
rRSA-256	1.8 KB	2.0 KB	3.1 ms	1.8 ms	128-bit PQ
CRYSTALS-Kyber	1.6 KB	2.4 KB	0.9 ms	1.1 ms	128-bit PQ
RSA-3072	3.1 KB	3.8 KB	12.5 ms	0.4 ms	128-bit classical

Table 1: Comparison with Post-Quantum Schemes

Analysis: Our schemes achieve comparable performance to established post-quantum systems while offering novel security assumptions.

4.3 Implementation Considerations

- **Floating Point Challenges:**
  - Use arbitrary-precision decimal arithmetic (GMP, MPFR libraries)
  - Standardize rounding modes across implementations
  - Include precision validation in communication protocol
- **Side-Channel Resistance:**
  - Constant-time function evaluation using Montgomery ladder
  - Blinded precision arithmetic to prevent timing attacks
  - Secure memory handling for function coefficients

5 Part 5: Cryptanalytic Evaluation

5.1 Known Attack Resistance

- **Lattice Attacks**: Function coefficients don't form exploitable lattice structures when properly randomized.

- **Algebraic Attacks**: Polynomial systems arising from known plaintexts require solving high-degree equations over large finite fields.
- **Timing Attacks**: Constant-time implementations prevent leakage of function parameters through execution timing.
- **Power Analysis**: Randomized computation order and value blinding protect against differential power analysis.

5.2 Quantum Cryptanalysis

- **Shor's Algorithm**: Not directly applicable — no pure integer factorization problem.
- **Grover's Algorithm**: Provides quadratic speedup for function space search and precision guessing:
  - Classical complexity $O(2^n) \to$ Quantum complexity $O(2^{n/2})$
  - Still exponentially hard for appropriately chosen parameters
- **Period Finding**: May apply to certain function classes — avoided by using non-periodic transcendental functions.

5.3 Novel Attack Scenarios

- **Function Interpolation Attack**: If degree n is small and evaluations $f(a), f(b)$ become known, polynomial interpolation becomes feasible. Mitigated by:
  - Using high-degree polynomials ($n \geq 8$)
  - Incorporating transcendental components
  - Adding random noise to function outputs
- **Precision Correlation Attack**: Patterns in truncated precision might leak information about full values. Mitigated by:
  - Cryptographically secure truncation protocols
  - Randomized precision adjustment
  - Hash-based precision validation

6 Conclusion and Future Work

The proposed fRSA and rRSA systems represent a paradigm shift in public-key cryptography, moving beyond reliance on single hardness assumptions toward multi-layered security architectures. By incorporating function-based transformations and precision-dependent security mechanisms, these systems offer:

1. **Proven quantum resistance** through novel computational challenges with formal security reductions

2. **Scalable security levels** adaptable to application requirements with concrete parameter recommendations
3. **Mathematical flexibility** in function selection and precision tuning with rigorous construction guidelines
4. **Practical implementability** through systematic variant structures and detailed synchronization protocols

Our formal analysis demonstrates that both systems achieve post-quantum security levels comparable to established schemes while maintaining computational efficiency. The precision-based security model offers a novel approach to creating exponentially difficult computational challenges that remain intractable even under quantum attack scenarios.

Experimental Results: Implementation prototypes achieve 128–256 bit post-quantum security with acceptable performance overhead (3–5x slower than classical RSA, comparable to other post-quantum schemes).

Future Research Directions:

1. **Standardization**: Formal submission to post-quantum cryptography standardization processes
2. **Advanced Function Classes**: Investigation of elliptic curve and modular form based transformations
3. **Zero-Knowledge Integration**: Protocols for proving function evaluation correctness without revealing functions
4. **Hardware Implementation**: FPGA and ASIC implementations for high-performance applications
5. **Formal Verification**: Machine-checked proofs of security properties using theorem provers

This work opens new research directions in adaptive cryptographic security and function-based transformation systems, providing a promising foundation for post-quantum cryptographic security that maintains the elegance and efficiency of prime-based mathematics while addressing the fundamental vulnerabilities of classical RSA.

7 Personal Note

This isn't quantum hard, this is potentially "quantum-impossible". For testing please head to my GitHub: https://github.com/Whoknowsme0nobody/fRSA-Official/